GATE ME 90 Days Challenge

Day 31/90 · Practice Set with Detailed Solutions

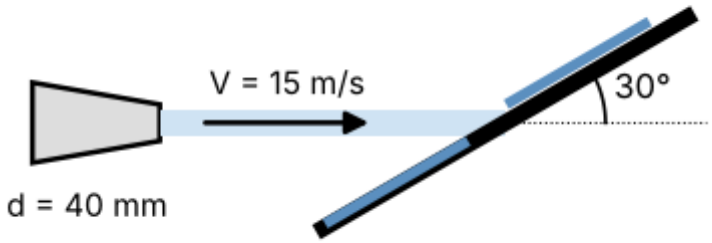
SUBJECT	TODAY'S TOPIC	COUNTDOWN
Fluid Mechanics	**Momentum, Flow Measurement & Dimensional Analysis**	**75 days to GATE**

How to use this set: 15 questions (5 MCQ, 10 NAT) on the momentum equation (jets on plates and vanes, pipe bends, nozzles), correction factors, flow measurement with venturimeters, orifice meters, orifices and pitot tubes, and dimensional analysis with model laws. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A water jet of 40 mm diameter and velocity 15 m/s strikes a fixed smooth flat plate inclined at 30° to the axis of the jet, as shown. The force exerted by the jet on the plate, normal to the plate, is



(A) 244.9 N
(B) 141.4 N
(C) 282.7 N
(D) 70.7 N

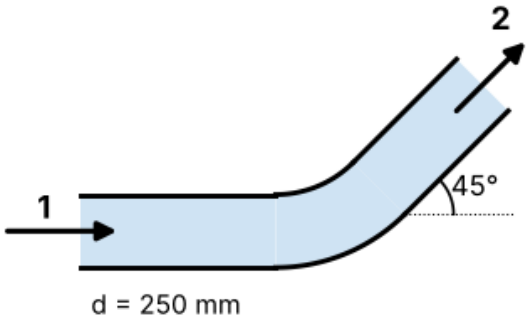
Q2 MCQ

A water jet strikes a series of flat plates mounted radially on the rim of a wheel (each plate normal to the jet). The maximum hydraulic efficiency attainable, and the plate speed u at which it occurs (V = jet velocity), are

(A) 50 %, u = V/3
(B) 100 %, u = V/2
(C) 50 %, u = V/2
(D) 29.6 %, u = V/3

Q3 NAT

Water flows at 0.15 m^3/s through the horizontal 45° bend of uniform 250 mm diameter shown. The pressure is 200 kPa (gauge) throughout the bend (losses neglected). The magnitude of the resultant force exerted by the water on the bend (in kN, round off to two decimal places) is _____



Q4 NAT

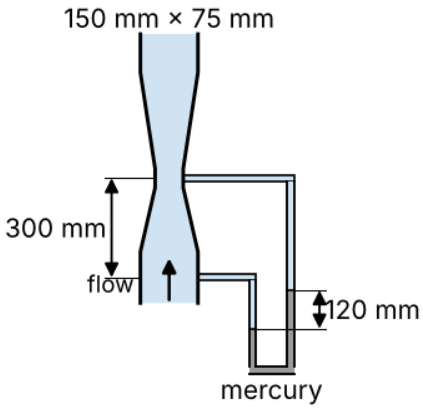
A nozzle at the end of an 80 mm diameter horizontal pipe discharges a 30 mm diameter water jet into the atmosphere at 10 litres per second. Neglecting losses, the force that the bolts at the flange must withstand (in N, round off to one decimal place) is _____

Q5 NAT

The velocity profile in a circular pipe of radius R is $u = U_{max}(1 - r/R)$. The momentum correction factor β for this profile (round off to two decimal places) is _____

Q6 NAT

A vertical venturimeter (inlet 150 mm, throat 75 mm diameter) carries water upward; the throat is 300 mm above the inlet section. The differential mercury manometer shown reads 120 mm. With C_d = 0.97, the discharge (in litres per second, round off to two decimal places) is _____

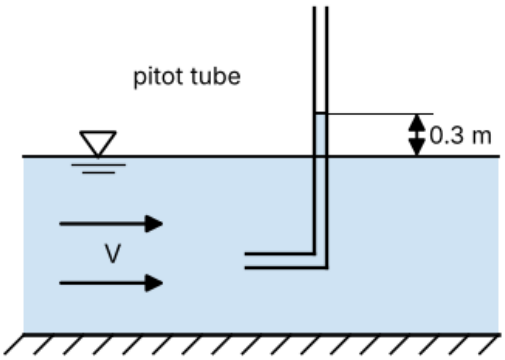


Q7 NAT

An orifice meter with a 100 mm diameter orifice is fitted in a 200 mm diameter horizontal water pipe. The pressure difference between the upstream tapping and the vena contracta tapping is 50 kPa. Taking C_d = 0.62, the discharge (in litres per second, round off to one decimal place) is _____

Q8 NAT

A pitot tube is placed in an open channel facing the flow, as shown. Water in the tube rises 0.3 m above the free surface of the stream. Taking the coefficient of the pitot tube as 0.98, the velocity of flow at the tip (in m/s, round off to two decimal places) is _____



Q9 MCQ

The wave-making resistance of a ship is to be predicted from tests on a scale model. The model law to be satisfied is

(A) Reynolds model law
(B) Froude model law
(C) Weber model law
(D) Mach model law

Q10 NAT

The thrust T of a marine propeller depends on its diameter D, speed of advance V, rotational speed N, fluid density ρ and viscosity μ. The number of dimensionless π terms is _____

Q11 MCQ

In very slow (creeping) flow the drag force F on a sphere depends only on the fluid viscosity μ, the velocity V and the diameter D. Dimensional analysis gives

(A) $F \propto \rho V^2 D^2$
(B) $F \propto \mu V^2 D$
(C) $F \propto \mu V D$
(D) $F \propto \mu V / D$

Q12 NAT

Oil (kinematic viscosity 1.2×10^{-5} m²/s) flows at 1.5 m/s in a 600 mm diameter pipe. A model test uses water (kinematic viscosity 1.0×10^{-6} m²/s) in a 150 mm pipe. For dynamic similarity, the velocity of water in the model (in m/s, round off to two decimal places) is _____

Q13 NAT

A 1 : 50 model of a ship, tested in sea water under Froude similarity, experiences a wave resistance of 2.5 N. The corresponding wave resistance of the prototype (in kN, round off to one decimal place) is _____

Q14 MCQ

For steady flow of a liquid through a pipeline, which statement about the energy grade line (EGL) and hydraulic grade line (HGL) is correct?

(A) The HGL always lies above the EGL by $V^2/2g$

(B) The EGL lies above the HGL by $V^2/2g$ and never rises in the flow direction unless a pump adds energy

(C) The HGL can never rise in the direction of flow

(D) The EGL and HGL coincide wherever the pipe is horizontal

Q15 NAT

Water flows out of a tank through a 25 mm diameter sharp-edged orifice under a constant head of 3 m. The measured discharge is 2.3 litres per second and the coefficient of contraction is 0.64. The coefficient of velocity (round off to three decimal places) is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(B) 141.4 N
Q2	MCQ	(C) 50 %, u = V/2
Q3	NAT	7.86 (accept 7.82 to 7.91)
Q4	NAT	371.5 (accept 369.3 to 373.7)
Q5	NAT	1.50 (accept 1.48 to 1.51)
Q6	NAT	24.11 (accept 23.96 to 24.25)
Q7	NAT	50.3 (accept 50.0 to 50.6)
Q8	NAT	2.38 (accept 2.36 to 2.39)
Q9	MCQ	(B) Froude model law
Q10	NAT	3
Q11	MCQ	(C) $F \propto \mu VD$
Q12	NAT	0.50 (accept 0.49 to 0.51)
Q13	NAT	312.5 (accept 311.6 to 313.4)
Q14	MCQ	(B) The EGL lies above the HGL by $V^2/2g$ and never rises in the flow direction unless a pump adds energy
Q15	NAT	0.954 (accept 0.950 to 0.958)

Detailed Solutions

Q1 **Answer: (B) 141.4 N**

Concept: On a smooth plate there is no friction along it, so only the jet's momentum component normal to the plate is destroyed: $F_n = \rho a V^2 \sin\theta$.

$a = \pi/4 \times 0.04^2 = 1.2566 \times 10^{-3}\ m^2$, $\rho a V^2 = 1000 \times 1.2566 \times 10^{-3} \times 15^2 = 282.7$ N

$F_n = 282.7 \times \sin 30° =$ **141.4 N**

Exam tip: Its component along the jet is $F_n \sin\theta = \rho a V^2 \sin^2\theta = 70.7$ N; read the question for which direction is asked.

Q2 **Answer: (C) 50 %, u = V/2**

Concept: With a series of plates the whole jet $\rho a V$ is used: work/s $= \rho a V(V - u)u$, input $= \rho a V^3/2$. $\eta = 2u(V - u)/V^2$; $d\eta/du = 0$ gives $u = V/2$ and $\eta_{max} =$ **50 %**.

Exam tip: For a **single** moving plate, mass striking/s is $\rho a(V - u)$; then the optimum is $u = V/3$ with $\eta = 8/27 = 29.6$ %. A Pelton bucket (180° deflection) can approach 100 %.

Q3 Answer: 7.86 (accept 7.82 to 7.91)

Concept: Momentum equation on a control volume: Σ(forces on fluid) = ρQ($V_{out} - V_{in}$), component by component.

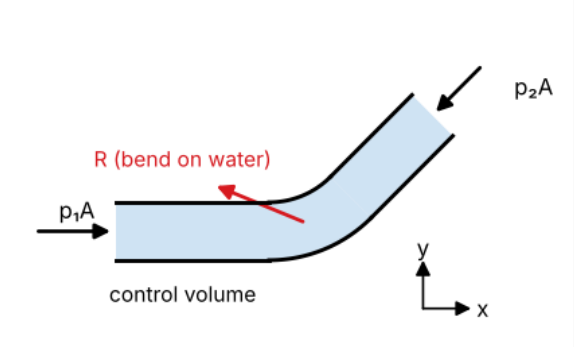
A = 0.04909 m², V = 0.15/0.04909 = 3.056 m/s; pA = 9.817 kN, ρQV = 0.458 kN, so pA + ρQV = 10.276 kN

x: F_x = (pA + ρQV)(1 − cos 45°) = 3.010 kN

y: F_y = (pA + ρQV) sin 45° = 7.266 kN

Resultant = $\sqrt{(F_x^2 + F_y^2)}$ = 2(pA + ρQV) sin 22.5° = **7.86 kN**. The bend pushes the water up and back (figure); the water pushes the bend equally outward, along the bisector of the bend angle.

Exam tip: For equal inlet and outlet pressure and area, R = 2(pA + ρQV) sin(θ/2). The pressure term usually dominates the momentum term.



Q4 Answer: 371.5 (accept 369.3 to 373.7)

Concept: Bernoulli gives the flange pressure; the momentum equation on the nozzle gives the force.

V_1 = 0.01/(5.0265 × 10⁻³) = 1.989 m/s, V_2 = 0.01/(7.0686 × 10⁻⁴) = 14.147 m/s

p_1 = ρ($V_2^2 - V_1^2$)/2 = 98.09 kPa (jet at atmospheric pressure), p_1A_1 = 493.1 N

$p_1A_1 - F = \rho Q(V_2 - V_1) \Longrightarrow F = 493.1 - 1000 \times 0.01 \times 12.158$ = **371.5 N** (the water pushes the nozzle downstream, trying to pull it off the pipe)

Exam tip: Forgetting the momentum term gives $p_1A_1 \approx 493$ N, a typical wrong option.

Q5 Answer: 1.50 (accept 1.48 to 1.51)

Concept: β = (1/AV^2)∫u^2 dA, with V the mean velocity.

V = (2/R^2)∫U_{max}(1 − r/R) r dr = 2U_{max}(1/2 − 1/3) = U_{max}/3

∫u^2 dA/A = 2U_{max}^2(1/2 − 2/3 + 1/4) = U_{max}^2/6

β = (U_{max}^2/6)/(U_{max}^2/9) = **1.50**

Exam tip: β = 1 for uniform flow, 4/3 for laminar (parabolic), about 1.02 for turbulent flow. The more non-uniform the profile, the larger β.

Q6 **Answer: 24.11 (accept 23.96 to 24.25)**

Concept: The manometer measures the piezometric head difference $h = x(S_m/S - 1)$; the elevation of the throat is already included in it.

$h = 0.12 \times (13.6 - 1) = 1.512$ m of water

$a_1 = 176.715 \times 10^{-4}$ m², $a_2 = 44.179 \times 10^{-4}$ m²

$Q = C_d\, a_1 a_2 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_2^2)} =$ **24.11 L/s**

Exam tip: The 300 mm height difference is a distractor: the reading of a differential manometer is the same whether the venturimeter is horizontal, inclined or vertical.

Q7 **Answer: 50.3 (accept 50.0 to 50.6)**

Concept: Same equation as the venturimeter, but C_d is much lower (0.6–0.65) because of the vena contracta and the large eddy losses.

$h = \Delta p/\rho g = 5.097$ m; $\sqrt{(2gh)} = \sqrt{(2\Delta p/\rho)} = 10.000$ m/s

$Q = 0.62 \times a_0 A_1 \times 10/\sqrt{(A_1^2 - a_0^2)} = 0.62 \times a_0 \times 10/\sqrt{(1 - (d/D)^4)} =$ **50.3 L/s**

Exam tip: The orifice meter is cheap and compact but has a high permanent pressure loss; the venturimeter ($C_d \approx 0.98$) loses very little.

Q8 **Answer: 2.38 (accept 2.36 to 2.39)**

Concept: The tip is a stagnation point: the extra rise above the free surface is the velocity head $V^2/2g$.

$V = C\sqrt{(2gh)} = 0.98 \times \sqrt{(2 \times 9.81 \times 0.3)} =$ **2.38 m/s**

Exam tip: In a closed pipe a pitot tube alone cannot work; you need the static pressure too (pitot-static tube or a differential manometer).

Q9 **Answer: (B) Froude model law**

Concept: Waves at a free surface are governed by gravity, so the Froude number $V/\sqrt{(gL)}$ must be the same for model and prototype.

Skin-friction resistance (Reynolds-dependent) is computed separately, because both Fr and Re cannot be matched with the same fluid.

Exam tip: Pipe flow, aircraft at low speed, submarines (fully submerged) → Reynolds; spillways, ships, open channels → Froude; high-speed aircraft → Mach; capillary flows → Weber.

Q10 **Answer: 3**

Concept: Buckingham π theorem: number of π terms = n − m.

n = 6 (T, D, V, N, ρ, μ), m = 3 (M, L, T) ⟹ **3** π terms, e.g. $T/(\rho V^2 D^2)$, ND/V, $\rho VD/\mu$.

Exam tip: Choose repeating variables (here ρ, V, D) that together contain M, L and T and are not themselves dimensionless as a group.

Q11 **Answer: (C) F ∝ μVD**

Concept: Write $F = k\mu^a V^b D^c$ and equate the powers of M, L and T.

$[MLT^{-2}] = [ML^{-1}T^{-1}]^a [LT^{-1}]^b [L]^c$: M: a = 1; T: $-a - b = -2 \Longrightarrow b = 1$; L: $-a + b + c = 1 \Longrightarrow c = 1$.

So $F = k\mu VD$; Stokes' solution gives $\mathbf{k = 3\pi}$, $F = 3\pi\mu VD$.

Exam tip: Stokes' law holds for Re < 1 and gives $C_D = 24/Re$.

Q12 **Answer: 0.50 (accept 0.49 to 0.51)**

Concept: Viscous pipe flow → equal Reynolds numbers: $V_m D_m/\nu_m = V_p D_p/\nu_p$.

$V_m = 1.5 \times (0.6/0.15) \times (1.0 \times 10^{-6}/1.2 \times 10^{-5}) = 1.5 \times 4 \times 0.0833 =$ **0.50 m/s**

Exam tip: Using a less viscous fluid lets a smaller model run at a modest speed; with the same fluid the model would need 4 × 1.5 = 6 m/s.

Q13 **Answer: 312.5 (accept 311.6 to 313.4)**

Concept: Under Froude law with the same fluid, $F_r = \rho_r L_r^2 V_r^2 = L_r^2 \times L_r = L_r^3$.

$F_p = 2.5 \times 50^3 = 312500$ N = **312.5 kN**

Exam tip: Froude scale ratios: velocity $L_r^{1/2}$, time $L_r^{1/2}$, discharge $L_r^{5/2}$, force L_r^3, power $L_r^{7/2}$.

Q14 **Answer: (B) The EGL lies above the HGL by $V^2/2g$ and never rises in the flow direction unless a pump adds energy**

Concept: EGL = $p/\rho g + z + V^2/2g$ (total head); HGL = $p/\rho g + z$ (piezometric head).

So the EGL is above the HGL by the velocity head. Losses make the EGL fall steadily; only a pump makes it jump up.

The HGL **can** rise, e.g. in a diverging section where velocity falls and pressure recovers.

Exam tip: Where the HGL drops below the pipe centreline the pressure is below atmospheric (as at a siphon summit).

Q15 **Answer: 0.954 (accept 0.950 to 0.958)**

Concept: $C_d = Q_{actual}/(a\sqrt{2gH})$ and $C_d = C_c C_v$.

$Q_{th} = 4.9087 \times 10^{-4} \times \sqrt{(2 \times 9.81 \times 3)} = 3.766$ L/s, so $C_d = 2.3/3.766 = 0.6107$

$C_v = C_d/C_c = 0.6107/0.64 =$ **0.954**

Exam tip: Typical sharp-edged orifice: $C_c \approx 0.62$–0.65, $C_v \approx 0.95$–0.99, $C_d \approx 0.6$–0.64.

What next?

Day 31 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 32/90: Fluid Mechanics, Viscous Flow & Boundary Layer. Comment **GATE90** on our reel to get all links.
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